10/9/26

EXPERIMENT-12A

TCP- ECHO CLIENT - SERVER

AIM: To implement an echo-client server ~~communication~~ Communication Using TCP socket.

ALGORITHM:

Server

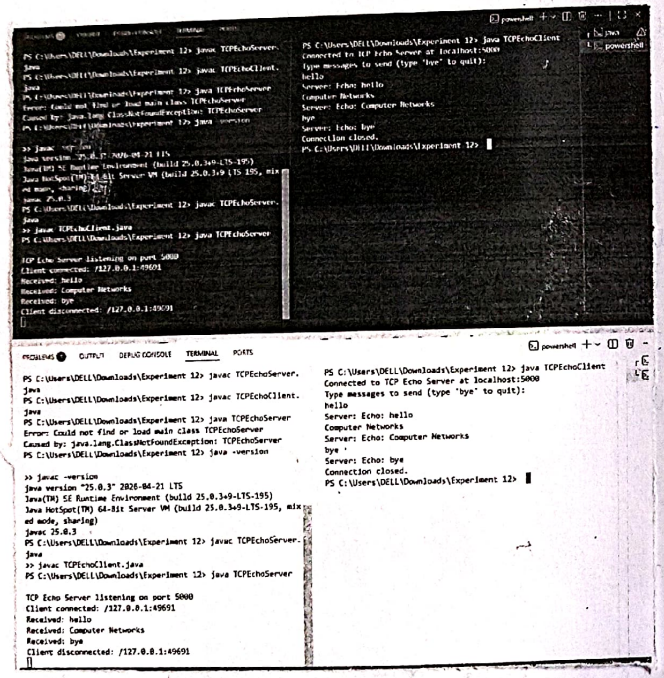
1. Create a TCP socket using SOCK-STREAM.
2. Bind the socket with port number.
3. Listen for client connection.
4. Accept client connection.
5. Receive message from client.
6. Send same message back to client.
7. Close Connection

Client:

1. Create a TCP Socket
2. Connect to server.
3. Get message from user
4. Send message to server
5. Receive echoed message
6. Display and close

TCP EchoClient.java

```
import java.io.*;
import java.net.*;
public class TCPEchoClient {
    public static void main(String[] args) throws IOException {
        String host = args.length > 0 ? args[0] : "localhost";
        int port = args.length > 1 ? Integer.parseInt(args[1]) :
                                                    5000;
```



```
try (Socket socket = new Socket(host, port);
     PrintWriter out = new PrintWriter(socket.getOutputStream(),
                                       autoFlush: true);
     BufferedReader in = new BufferedReader(new InputStream
                                            Reader(socket.getInputStream()));
     BufferedReader stdIn = new BufferedReader(new InputStream
                                               Reader(System.in))) {

    System.out.println("Connected to TCP Echo Server at " +
                       host + ":" + port);
    System.out.println(x: "Type messages to send (type 'bye'
      to quit):");

    String userInput;
    while ((userInput = stdIn.readLine()) != null) {
        out.println(userInput);
        String response = in.readLine();
        System.out.println("Server: " + response)
        if (userInput.equalsIgnoreCase(anotherString:
                "bye")) {
            break;
        }
    }

    System.out.println(x: "Connection closed.");
}
```

TCP EchoServer.java

```
import java.io.*;
import java.net.*;
public class TCPEchoServer {
    public static void main (String [] args) throws IOException {
        int port = args.length > 0 ? Integer.parseInt (args[0]): 5000;
        try(ServerSocket serverSocket = new ServerSocket (port)) {
            System.out.println("TCP Echo Server listening on Port " +
                    port);

            while (true) {
                try (Socket clientSocket = serverSocket.accept()){
                    System.out.println ("Client connected" +
                        Client Sock.getRemoteSocketAddress());

                    BufferedReader in = new B
```

RESULT:

Thus, the TCP Echo client Server application was succesfully implemented